

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Kanigasi M

Code of Conduct:

I Kanigasi/192565010 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Kanigasi M
Signature of the student:

16
/ 20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Question 1

A Smart home automation System Controls lighting, temperature and Security by responding to user behaviour and environmental conditions. Different types of Intelligent agents perform different types of roles in the System.

1) Simple reflex Agent

→ responds only to the current input using predefined rules

Example: Turns on lights when motion is detected

→ Fast but cannot remember past events

2) model-based Agent

→ responds based on previous actions and surroundings

Example: ~~turns on~~ remember the surrounding and controls

→ more complex ~~to~~ and remember the surrounding which makes the control easier

3) Goal based Agent

- Makes decisions to achieve a specific Goal
- Example : Maintains a comfortable home temperature and ensures security
- Can plan before taking action

4) Utility based Agents

- Makes decisions based on what is being used
- Example : Works based on the preferences
- remembers both previous preferences and surrounding

Best Approach:

A hybrid of Model - Base and Utility Based Agents is the most effective because it remembers previous information and preferences for and would be most effective for ensuring comfort, energy efficiency and security

Q
A robot placed in an unknown maze must find the exit without knowing the layout. uninformed search algorithms like Uniform Cost Search (UCS) and iterative Deepening Search (IDS) help solve this problem.

1) Uniform Cost Search (UCS)

UCS expands the path with the lowest total cost first until the goal is reached.

- Finds the least-cost path.

- Suitable when movement costs are different.

- Advantages: Complete and optimal.

- Disadvantages: Require more memory and takes longer for larger problems.

2) Iterative Deepening Search (IDS)

IDS searches for a path until the best one is found that the goal can be reached

→ Finds the path slow and steady

→ Suitable when movement costs are similar

→ Advantage: steady and useful

→ disadvantage: can be slow and require a lot of time

Best Approach:

The Best Approach for this search would be Goal based uniform cost search as it makes decision to achieve a specific goal and expands the path in lowest total cost

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand